Online News Popularity

What drives article shares, and can virality be predicted?

Rajashree Guduru | M.S. Business Analytics, Sacred Heart University

0.73	0.66	0.529
ROC-AUC (gradient boosting)	Test accuracy	AUC from headline embeddings alone

1. Background and question

Which article characteristics are associated with more shares, can we predict whether an article will be popular, and does headline text add signal?

Mashable publishes thousands of articles, and shares drive reach and advertising revenue. Editors control a handful of levers (topic and channel, keywords, length, media, publishing day), so knowing which ones are linked to shares helps decide where to spend effort.

2. Data

- UCI Online News Popularity dataset: 39,644 Mashable articles (2013-2014)
- 58 predictive features: content length, links, images and videos, keyword statistics, publishing day, channel, LDA topic scores, sentiment and subjectivity
- Target: shares (median 1,400, heavily right-skewed: skewness 34); popular = more than 1,400 shares (49% of articles)

3. Method

- Exploratory analysis of share distribution, channel and weekday effects
- Driver analysis: OLS regression with controls on log(1 + shares), standardised features, redundant dummies removed to avoid collinearity
- Prediction: majority baseline, logistic regression, gradient boosting on a stratified 80/20 split
- Text experiment: headlines extracted from URL slugs and embedded with TF-IDF (unigrams + bigrams), tested alone, combined with numeric features, and as an out-of-fold score in gradient boosting
- LLM embeddings: each headline embedded with the all-MiniLM-L6-v2 sentence-transformer (384 dimensions), tested alone and as an out-of-fold score in gradient boosting; K-means on the embeddings grouped headlines into 12 topic clusters
- Permutation importance for the best model

4. Project walkthrough

The problem. Two related questions: which article characteristics go with more shares (explanation), and can we predict whether an article will be popular before it is published (prediction)?

The data. 39,644 articles with 58 pre-computed features. Shares are extremely skewed: the median is 1,400 but one article reached 843,300. I defined 'popular' as more than 1,400 shares, which splits articles roughly in half.

Explaining shares. For the explanation question I used OLS regression with controls on log shares. Taking the log stops a few viral articles from dominating the fit. Entering all features at once means each coefficient is the effect of one feature holding the others constant: the 'controls'.

Predicting popularity. For prediction I compared a baseline (0.51 accuracy), logistic regression and gradient boosting. Gradient boosting builds many small decision trees, each correcting the previous ones' mistakes, so it captures non-linear effects; it reached 0.73 ROC-AUC.

Testing headline text. The dataset has no article text, but each URL contains the headline. I tested two text representations: TF-IDF word features, and LLM sentence embeddings from the all-MiniLM-L6-v2 transformer, which capture meaning rather than exact words. Neither improved the model: embeddings alone scored AUC 0.529, and adding them left gradient boosting at 0.728.

Topic clusters. Clustering the embeddings grouped headlines by meaning into 12 topics. Facebook / social and photo stories had the highest median shares; hard news and sports the lowest popularity rates. So embeddings were more useful for organising content than for prediction.

The findings. Keyword strength (how well articles with the same keywords perform) is the biggest driver. Weekend articles earn about 20-25% more shares than weekday ones, and Social Media and Tech outperform Entertainment, Business and World.

Why it matters. The levers that matter are what you write about and when you publish, not how you word the headline, and a negative result like the headline test is still useful because it tells editors where not to spend effort.

5. Results

Model	Accuracy	ROC-AUC
Majority-class baseline	0.507	0.500
Logistic Regression (numeric)	0.651	0.701
Gradient Boosting (numeric)	0.664	0.728
Headline TF-IDF only	0.519	0.526
Headline LLM embeddings only (MiniLM)	0.522	0.529
Gradient Boosting + headline TF-IDF score	0.667	0.728
Gradient Boosting + LLM embedding score	0.668	0.728

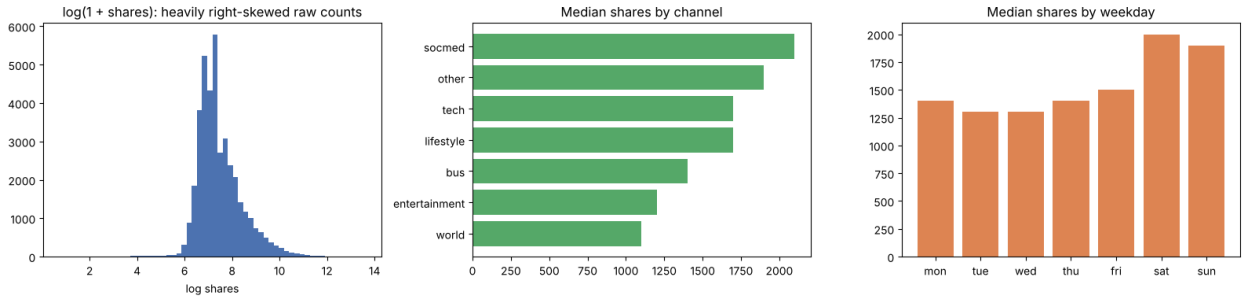


Figure: Share distribution (log scale), median shares by channel and by weekday

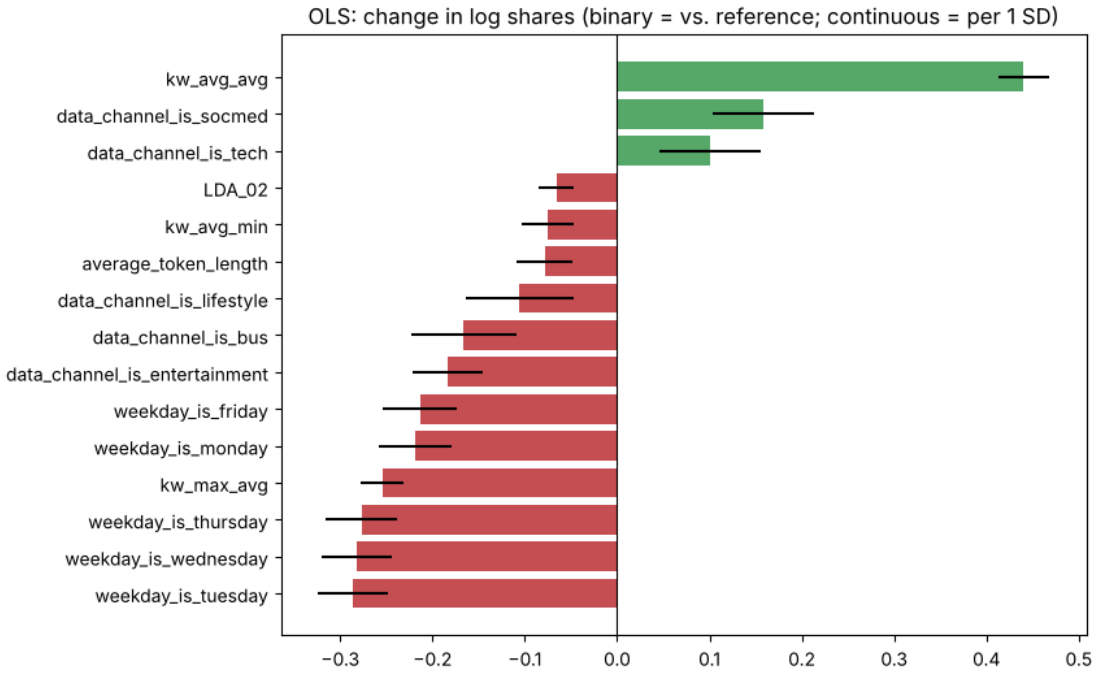


Figure: OLS with controls: strongest associations with log shares (95% intervals)

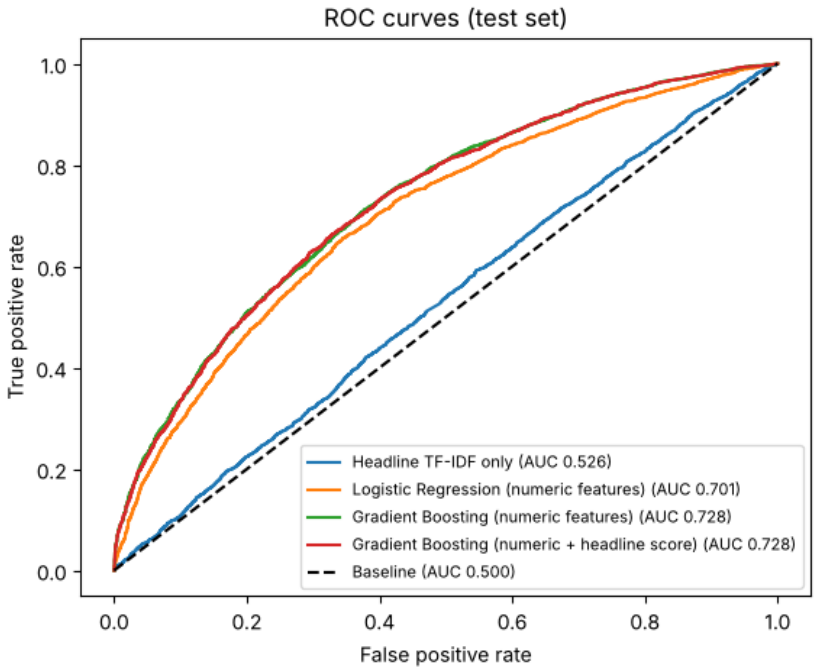


Figure: ROC curves on the test set

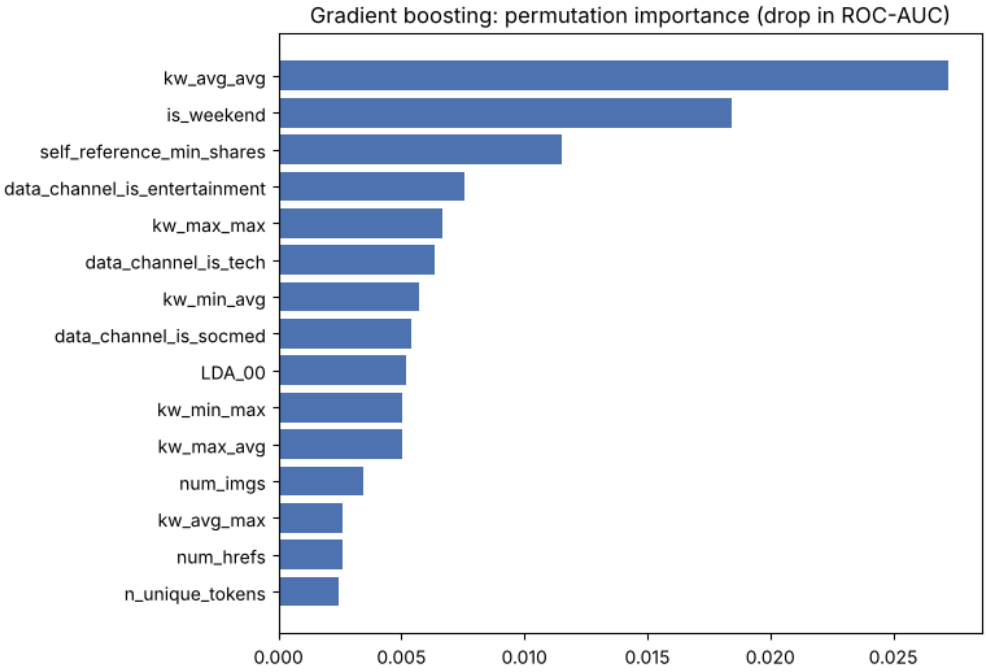


Figure: Gradient boosting permutation importance

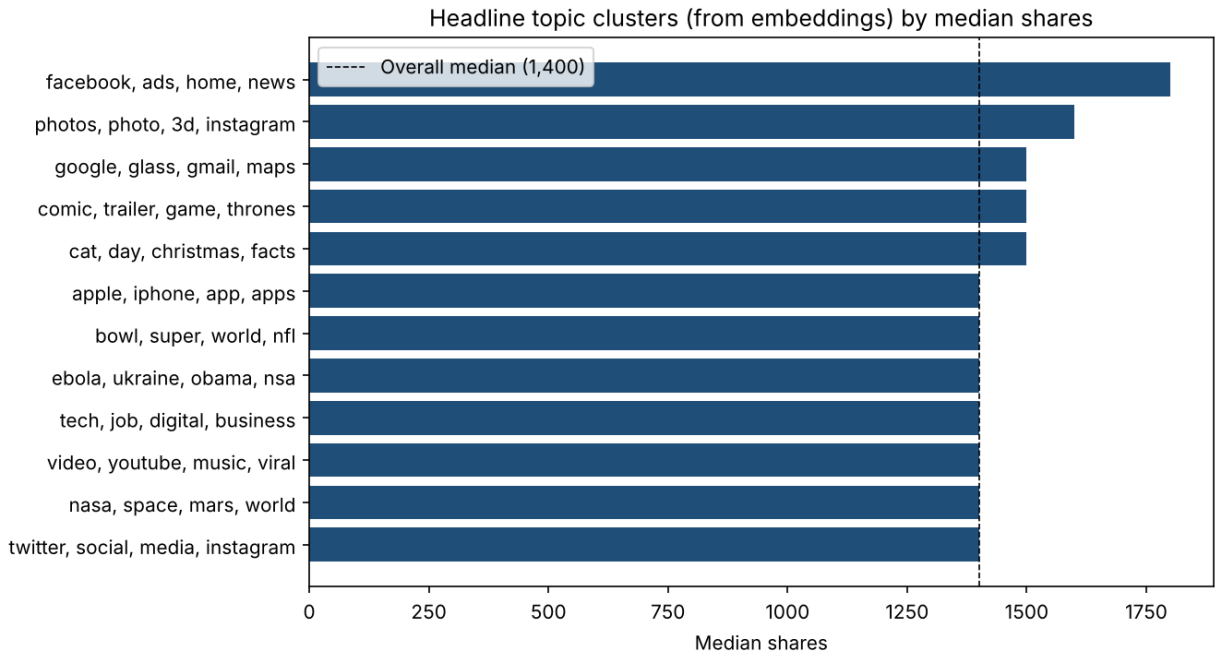


Figure: Headline topic clusters found with LLM sentence embeddings, by median shares

6. Key findings

- Gradient boosting lifted ROC-AUC from 0.50 to 0.73 and accuracy from 0.51 to 0.66, beating logistic regression by capturing non-linear effects.
- Headline text added no signal with either representation: TF-IDF (AUC 0.526) or LLM sentence embeddings (0.529). Adding the embedding score left gradient boosting at 0.728.
- Keyword strength is the largest driver: articles whose keywords historically earn more shares get substantially more shares.
- Weekend publishing helps: Monday-Friday articles get about 20-25% fewer shares than Sunday articles (median 1,300-1,500 vs. 1,900-2,000 on weekends).
- Social Media and Tech channels outperform; Entertainment, Business and World underperform.
- LLM embedding topic clusters: Facebook / ads / social stories have the highest median shares (1,800; 58% popular) and photo / Instagram stories follow (1,600); hard-news (Ebola, Ukraine, NSA) and sports clusters have the lowest popularity rates (46-47%).

7. Why these choices

Log transform for regression. Raw shares have skewness 34. Without the log, a handful of viral articles would decide every coefficient.

Explanation and prediction as separate steps. OLS gives interpretable effect sizes; gradient boosting gives the best predictions. Using both answers 'why' and 'how well'.

ROC-AUC alongside accuracy. AUC measures ranking quality across all thresholds, which matters if an editor wants the top 10% most promising articles rather than a yes/no label.

Out-of-fold headline scores. When adding the headline score to gradient boosting, I generated it with cross-validation so the model never saw scores computed from its own training labels (avoiding leakage).

Two text representations. Testing both TF-IDF and LLM embeddings made the 'no lift' conclusion robust: if meaning-aware embeddings also add nothing, headline wording genuinely carries little signal here.

8. Key terms explained

OLS with controls	Linear regression with many features entered together, so each coefficient is estimated holding the others constant.
Log shares	Natural log of (1 + shares). A coefficient of -0.25 means roughly 22% fewer shares (e^-0.25 = 0.78).
Gradient boosting	An ensemble of small decision trees built in sequence, each fixing the errors of the last.
TF-IDF	Text features that count words, down-weighting words that appear in almost every headline.
Permutation importance	How much the model's AUC drops when one feature's values are shuffled; bigger drop = more important feature.
kw_avg_avg	The average shares of past articles using the same keywords: a measure of how popular the topic already is.
Sentence embeddings	A pretrained transformer maps each headline to a 384-number vector so that headlines with similar meaning are close together, even with different words.

Tools and skills

Python, pandas, scikit-learn, OLS regression, gradient boosting, sentence-transformers (LLM embeddings), TF-IDF, K-means clustering, permutation importance, matplotlib

Code: online_news_popularity.ipynb in this folder reproduces every number in this report.